Decision rules rather than representations drive external failures of diabetic retinopathy grading models

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Abstract

Placeholder.

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Introduction

Diabetic retinopathy (DR) affects roughly one in five people with diabetes, an estimated 103 million adults in 2020 and a projected 161 million by 2045, and it remains a leading cause of preventable blindness in the working-age population [1]. Diabetic macular oedema (DME), its most common mechanism of central vision loss, affects about one in fifteen [2]. Deep learning systems that grade fundus photographs reached specialist-level agreement in development studies a decade ago [3,4], and autonomous screening devices have since been authorised on the strength of prospective pivotal trials [5,6]. Yet the same class of system behaves very differently once it leaves the population it was built on. In a head-to-head evaluation of seven commercial algorithms on the same real-world images, sensitivity for referable disease ranged from 51% to 86% [7]; three commercial algorithms applied to the same eyes against the same reference standard in Indian public-health clinics returned specificities between 14% and 96% [8].

When a model fails on a new population, the failure is usually read as a statement about what the network has learned. A recent frozen-encoder benchmark found that

two of three encoders, including the retinal foundation model RETFound, did not identify a single one of 270 mild cases in an external corpus (grade-1 F1 = 0.000), and attributed the collapse to the encoders' representations [9]. RETFound itself was shown to be superior using macro-averaged, threshold-free metrics [10], which cannot reveal a per-grade failure of this kind. Comparisons between architectures, heads and pretraining schemes are reported in the same way: as differences in what a model represents.

An ordinal grading model does not output a grade, however. It outputs scores that a decision rule turns into a grade, and the rule is almost never examined. Three different things are routinely conflated: *probability calibration*, whether predicted probabilities match observed frequencies [11,12]; the *decision rule*, the cut-points or referral threshold that map scores to a grade or a referral; and the *representation*, what the encoder has learned. The common default, a threshold of 0.5 on every ordinal output or an argmax, is a hyper-parameter that has been chosen implicitly rather than one that has been avoided. If a model ranks images correctly but its cut-points sit in the wrong place for a new population, it will look blind to a grade it in fact separates, and two models that differ mainly in where their defaults happen to fall will look as if they differ in what they learned. Clinical prediction research has long treated recalibration and model updating as a routine step before deployment [13,14]; DR grading studies rarely test whether their headline failures and gains survive it.

Here we test that alternative directly. We trained and archived 13 grading configurations on a pooled development set of 2,260 images, evaluated them on three external directions (3,662 APTOS images, 12,522 DDR images from 147 Chinese hospitals, and the reverse direction, APTOS to Messidor-2 and IDRiD), and asked, for every reported failure and every reported difference, whether it survives *matched calibration*: refitting the decision rule of every model by the same procedure before comparing them. Every training run and every hypothesis was pre-registered with its falsifying outcome before launch [15], and we report the interventions that failed with the same weight as those that worked.

We find that the decision rule, not the representation, explains the external collapse of the mild grade and most apparent differences between models. Moving cut-points alone, with no retraining, raised external mild-DR recall several-fold; the choice among deployable decision rules changed clinically decisive metrics more than the choice among models; quadratic weighted kappa (QWK), the field's standard summary, was nearly flat across operating points that differed by tens of points in referral sensitivity; and models that differed only by random seed produced per-grade "effects" of the size that the literature attributes to architecture. Operating points fitted on development data did not transfer to new populations, while about 200 locally labelled images were enough to recalibrate safely. By contrast, eight of nine pre-registered interventions on the representation itself were falsified.

Methods

Study design and pre-registration

This is a retrospective modelling study on public, de-identified fundus photographs. It was conducted under an evaluation protocol frozen before the first training run (unit of independence, held-out sets, model selection on the development pool only, metrics and uncertainty), and every experiment was pre-registered: its hypothesis, control and falsifying outcome were committed to the version-controlled repository before the run was launched, and the commit history is the record [15]. The nine interventions on the model and the nine outcomes O1–O9 of the journal experiments (seed replicates, the reverse direction, the second external corpus and the foundation-model completion) were registered in this way; a training run cannot be launched by our kernel builder without a recorded hypothesis. Reporting follows TRIPOD+AI [16] and CLAIM [17] where applicable (Supplementary Table S1).

Datasets

The *development pool* combined IDRiD [18,19] (516 images; 5-grade International Clinical DR scale [20] and a 3-grade risk-of-DME label defined by the distance of the nearest hard exudate to the macula) and Messidor-2 [21,22] (1,744 gradable images with adjudicated DR grades and a binary referable-DME label [23]), 2,260 images in total (Table **??**). IDRiD's official train/test partition was pooled, because a 103-image test split cannot resolve the effects studied here (half-width of a 95% interval on accuracy $\approx$6 points). The *external* DR sets were APTOS 2019 (3,662 images from Aravind Eye Hospital, India; single-grader labels) and DDR [24] (12,522 gradable images from 147 hospitals in 23 Chinese provinces, graded by seven graders; images distributed at 512 pixels). Neither was used for training, model selection or threshold fitting, and both were held out from the first run. The EyePACS 2015 training set (35,126 images, single-grader labels) was used only to pretrain the DR head. In the reverse-direction experiment APTOS was the training corpus and Messidor-2 and IDRiD the external sets. Per-image identity checks found no image of any external set in any training set. No patient identifiers are published for IDRiD, Messidor-2 or APTOS; the unit of resampling is therefore the image, and near-duplicate images were merged into one group by a 256-bit difference hash calibrated on fundus images (distinct pairs $\geq$28 bits apart, re-encodings of one image $\leq$8; threshold 16).

Labels

DR labels are the 5-grade international scale in all corpora (0 no DR, 1 mild, 2 moderate, 3 severe non-proliferative, 4 proliferative); referable DR is grade $\geq$2. DME is graded 0–2 in IDRiD (none; exudates more than one disc diameter from the macula; within one disc diameter). Messidor-2's binary referable-DME label is, by its definition, exactly IDRiD's grade 2; a non-referable Messidor-2 image is grade 0 *or* 1. In threshold form this partial label supervises the second DME threshold and leaves the first unsupervised for that image, so no label is imputed. The 3-grade DME metrics

are computed on IDRiD only (516 images, 51 of them in the middle grade), always beside the majority-class floor (47.1%), and without gating on DR.

Model and training

All models share one design: a convolutional (or transformer) backbone, global average pooling, and two ordinal threshold heads (hidden width 256, dropout 0.4) that output $K-1$ logits for $P(y>k)$, four for DR and two for DME[25]. The loss is $0.6\,\mathcal{L}_{\mathrm{DR}}+0.4\,\mathcal{L}_{\mathrm{DME}}$, each a masked binary cross-entropy over the thresholds with label smoothing 0.05. Images were cropped to the retinal disc, cached at a long side of 560 pixels and resized to the training resolution (448 pixels unless stated), with unrestricted rotation, scaling (0.88–1.12), translation (±4%), horizontal and vertical flips and mild photometric jitter; a sampler balanced the DR grades. Training used AdamW (backbone learning rate 10^{-4}, heads 10^{-3}, weight decay 10^{-4}), a 5% linear warm-up followed by cosine decay to 2% of the peak, gradient clipping at 1.0, mixed precision, and an exponential moving average of the weights (decay 0.999) for evaluation. The reference configuration (E08) is DenseNet121[26] pretrained on ImageNet, then on EyePACS for 5 epochs, then fine-tuned for 40 epochs with batch 16. The selected model (E11FULL) replaces the backbone with EfficientNet-B3[27] (30 epochs). Every other configuration changes exactly one factor relative to its control (Supplementary Table S2): resolution (224, 560 and native 640 pixels), a macula-centred crop for the DME head driven by a fovea localiser validated before use (median error 0.196 disc diameters), linear-probe-then-fine-tune[28], CORAL and CORN ordinal heads[25,29], ConvNeXt-tiny[30], and the RETFound colour-fundus ViT-L/16 encoder[10] (224 pixels, linear probe for 5 epochs then fine-tuning at 10^{-5}). At the end of each fold the epoch with the highest $0.6\,\mathrm{QWK}_{\mathrm{DR}}+0.4\,\mathrm{QWK}_{\mathrm{DME}}$ on the held-out fold was kept, and its predictions were averaged over the four dihedral flips (test-time augmentation). External predictions average the logits of the five fold models.

Cross-validation and model selection

All development results are out-of-fold over the full pool: a 5-fold split, stratified by corpus and DR grade and grouped by near-duplicate cluster, was built once, fingerprinted and committed before the first run, and every model loads it from that file. Configurations were ranked by development QWK at matched calibration (below) and the selected model was declared in the repository before any external number was examined; no external result was used to choose a model, a threshold or a hyperparameter. Because the checkpoint epoch was chosen on the held-out fold, development estimates carry a small selection optimism. We estimated it by re-selecting each fold's epoch from the *other* folds' validation curves: across nine 5-fold runs the median optimism was +0.0025 DR QWK (range −0.0008 to +0.0159). External estimates are not affected.

Decision rules and matched calibration

For an ordinal model with threshold probabilities $p_k = P(y>k)$ the expected grade is $s=\sum_k p_k$. We compare five decision rules, fixed before the journal experiments.

R1 is the shipped rule: grade = the number of leading thresholds with $p_k > 0.5$ (made rank-consistent with a cumulative minimum). **R2** places four cut-points on s to maximise QWK on the development out-of-fold predictions. **R3** is R2 with the grade-1/2 cut moved to the most specific value that gives a development referable sensitivity of at least 0.90. R1–R3 are deployable: none uses external labels. **R4** and **R4m** fit the cut-points on the external labels themselves, maximising QWK or macro-averaged recall, by two-fold cross-fitting (each half's cut-points are fitted on the other half); they are diagnostic upper bounds on what recalibration with local labels can recover and are never reported as performance. Cut-points are found by coordinate ascent (eight rounds; 60 candidates per cut on development data and 80 on external data, reproducing the archived analyses exactly; the grid alone moves a matched QWK by at most 0.0033).

Matched calibration is the comparison rule applied throughout: no difference between two models is attributed to representation unless it survives refitting both models' cut-points by the same cross-fitted procedure (on the development pool, fold f's cut-points come from the other four folds). The rule is only evidence if it can return either answer; its record in this study is reported in Table **??**.

Metrics and statistical analysis

The primary grading metric is QWK[31]; the operationally primary pair is referable-DR sensitivity and specificity, never quoted alone; per-grade recall is reported beside every headline, and macro-averaged recall is used as a diagnostic, never as an objective (optimising it costs referral sensitivity; see Results). We also report Cohen's unweighted κ, accuracy and macro-F1, and, for the referral probability $P(\text{grade} \geq 2) = p_1$, the area under the ROC curve, the Brier score and the expected calibration error (ECE, ten equal-width bins) together, since ECE alone can reward an uninformative predictor[9]. Intervals are 95% percentile intervals from 2,000 bootstrap resamples of groups; differences between models are paired bootstraps over the same groups, and a difference whose interval contains zero is reported as indistinguishable, not ranked. No correction for multiple comparisons is applied; per-grade claims are therefore treated as credible only when they replicate across conditions.

Recalibration sample size

To quantify how many locally labelled images recalibration needs, we drew $n \in \{25, 50, 100, 200, 400\}$ external images at random (200 draws per size), fitted cut-points on those n alone (maximising macro-recall), and evaluated on the remaining images, so that every estimate is out of sample with respect to its cut-points, as in deployment. The probability of harm is the fraction of draws in which macro-recall fell below that of the shipped rule on the same held-out images.

Seed replicates, reverse direction and second external corpus

Seeds 1–4 of the reference configuration were trained with every other setting identical (E08 is seed 0; consumption manifests confirm the same files were read), and each was scored on APTOS and DDR. The reverse-direction model used the reference

recipe trained on APTOS with a 5-fold split built by the same code before training (near-duplicates merged), and was scored on Messidor-2 and IDRiD. The archived development models were re-scored on DDR (and those without archived APTOS predictions on APTOS) from their stored fold weights, with the same inference recipe.

Provenance controls

Each run records, from inside the process that read the disk, a consumption manifest (number of images per corpus and a hash over the resolved paths); runs whose manifests differ cannot be combined or compared without the difference being printed into the output. Every number in this article is generated from archived artefacts into a ledger, and the manuscript is compiled from that ledger, so no number in the text is typed by hand.

Use of AI assistance

An AI system (Claude, Anthropic) assisted with analysis code, the execution of the pre-registered journal experiments and drafting of the manuscript under the direction of the authors, who verified the analyses, the numbers against their generated sources, and every cited claim against its primary source, and take full responsibility for the content.

Data availability

All images are from public datasets: IDRiD (IEEE DataPort), Messidor-2 (ADCIS, with adjudicated grades from Krause *et al.*[23]), APTOS 2019 and EyePACS 2015 (Kaggle) and DDR[24]. Per-image out-of-fold and external predictions for every model, the frozen split, the pre-registration records and every generated table are available in the project repository (https://github.com/Ahmadrezanourozii/Automatic-detection-of-diabetic-retinopathy-and-grading-of-diabetic-macular-edema-using-CNN).

Code availability

All code (training, evaluation, decision-rule analyses, figure and number generation) is available in the same repository; each run is reproducible from the commit hash recorded in its log.

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